



Filter Element Datasheet - Lube Oil

Technical ratings, differential pressure guidance, and compatibility chart for turbine lube oil filter elements supplied by Keshav Enterprises.

Reference standards: ISO 16889 (multi-pass method for evaluating fine filter element performance — the basis of the Beta ratio rating) and ISO 4406:2021 (fluid cleanliness classification by particle count). This datasheet summarizes common industry practice under these standards; it does not reproduce their full text.

1. General Specification Range

Parameter	Typical Range	Notes
Micron Rating	3, 6, 10, 25 micron (Beta 1000 grade available)	Per system cleanliness target
Media Type	Glass-fibre / cellulose / stainless wire mesh	Cellulose not for high-moisture service
Max. Operating Temperature	Up to 100 deg C (continuous)	Higher on request for special media
Max. Differential Pressure	5 - 10 bar (element collapse rating)	See individual element datasheet
Recommended Change-Out DP	2.0 - 2.5 bar differential	Fit DP gauge / switch on housing
End Cap Material	Zinc-plated steel / stainless steel	Stainless for corrosive environments
Seal Material	Nitrile (NBR) standard; Viton on request	Match to oil type and temperature

2. Differential Pressure vs. Flow (Guideline Curve Points)

Flow Rate (LPM)	Clean Element DP (bar)	50% Loaded DP (bar)	Change-Out DP (bar)
50	0.15	0.55	2.0
100	0.30	0.85	2.0
150	0.45	1.15	2.2
200	0.65	1.50	2.5

Indicative values for a standard 10-micron glass-fibre element in a single cartridge housing. Actual DP depends on oil viscosity, temperature, and element loading — refer to the housing manufacturer's curve for precise sizing.

3. OEM / Make Compatibility Cross-Reference

OEM / Brand	Typical Housing Series	Keshav Cross-Reference Available
HYDAC	Return line, pressure line filters	Yes - direct dimensional match
PALL	Duplex & simplex turbine filters	Yes - direct dimensional match
BHEL	OEM turbine lube oil filters	Yes - reverse-engineered spec
Triveni / Siemens / BHEL housings	Various duplex housings	Yes - site measurement based
Bhagwati	Replacement oil filter elements	Yes - direct replacement



4. Beta Ratio & Filtration Efficiency (ISO 16889)

The Beta ratio (Beta-x) is the count of particles larger than size x upstream of the filter divided by the count downstream, measured on a standardized multi-pass rig per ISO 16889. It is a far more precise measure of real filtration performance than a plain 'micron rating' claim, since nominal micron ratings are not standardized between manufacturers. Efficiency is calculated as $(\text{Beta} - 1) / \text{Beta} \times 100$.

Beta Ratio	Capture Efficiency at Rated Micron	Typical Use
Beta 2	50%	Coarse strainer / suction screen
Beta 10	90%	Return line, general protection
Beta 20	95%	Return line, moderate protection
Beta 75	98.7%	Pressure line, standard protection
Beta 200	99.5%	Pressure line, high protection (recommended minimum for turbine lube oil)
Beta 1000	99.9%	High-precision / EHC servo systems
Beta 2000	99.95%	Critical high-precision systems

When specifying a replacement element, ask for the Beta ratio at the required micron size, not just a micron number — a Beta 200 (or higher) rating at the stated micron size is generally recommended for turbine lube oil service, tightening to Beta 1000+ for electro-hydraulic control (EHC) and servo-valve circuits where a target ISO 4406 code of 16/14/11 or cleaner applies.

5. Ordering Information

To receive a like-for-like or cross-referenced quotation, please share: (1) existing element part number or a physical sample, (2) housing make and model, (3) outer diameter, inner diameter, and length (mm), (4) micron rating required, and (5) end-cap / seal configuration. Keshav Enterprises reverse-engineers filter elements where OEM part numbers are obsolete or unavailable in India.

Values in this datasheet are representative of standard turbine lube oil filter elements and are provided for reference and sizing guidance. Final specifications are confirmed at the time of order based on the exact housing and application.